

Biniyam Aschalew Tolera

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EDUCATION

KAIST, Daejeon, Republic of Korea

M.S. in Electrical Engineering

Advisor: [Jaemin Yoo](#)

Sep. 2025 – Jul. 2027 (expected)

Cumulative GPA: 3.92/4.3

KAIST, Daejeon, Republic of Korea

B.S. in Computer Science, 🏆 magna cum laude

Minor in Industrial Engineering

Mar. 2021 – Aug. 2025

Cumulative GPA: 3.87/4.3

RESEARCH INTERESTS

Robust and transferable machine learning; graph representation learning; foundation models for structured data.

RESEARCH EXPERIENCE

Data AI Lab, KAIST

Jun. 2024 – Present

M.S. Student (Sep. 2025 – Present); Research Intern (Jun. 2024 – Aug. 2025)

Advisor: Prof. Jaemin Yoo

- **OPAL: Structural Domain Shift in Graphs [M1]**. Developed OPAL, an unsupervised graph domain adaptation method that learns graph filters with aligned response distributions to improve transfer under structural shift.
- **Scalable In-Context Learning with Tabular Foundation Models**. Investigating tabular foundation models as in-context learners for tabular, time-series, and graph data using suitable encoders; developing learned context selection to reduce the quadratic computational cost of large context sets.
- **PULSE: Parameter-Efficient Graph Collaborative Filtering [C1]**. Implemented and evaluated model ablations to assess the contribution of individual components to recommendation performance.

Networking and Mobile Systems Laboratory, KAIST

Jul. 2023 – Jun. 2024

Research Intern | Advisor: Prof. Sung-Ju Lee

- **By My Eyes: Visual Prompting for Sensor Data [C3]**. Contributed to visual-prompting ideation and led the full experimental evaluation, including sensor-data pipelines, baseline implementation, and studies of prompt design and sequence-length effects.
- **SelfReplay: Adaptation of Self-Supervised Sensing Models [C2]**. Contributed to ideation for cross-domain and user adaptation; implemented baselines and designed and ran evaluations of self-supervised meta-learning.

Complexity and Real Computation Laboratory, KAIST

Mar. 2024 – Aug. 2024

Individual Study | Advisor: Prof. Martin Ziegler

- Designed experiments using the Hofstadter test to evaluate LLM agents in a virtual world.

Institute for Basic Science (IBS)

Dec. 2022 – Feb. 2023

Individual Study | Advisor: Prof. Meeyoung Cha

- Analyzed satellite imagery for Venusian climate prediction and evaluated deep learning models for weather classification.

INTERNSHIP EXPERIENCE

Bastion Robotics, Inc.

Dec. 2023 – Feb. 2024

AI Intern, Seoul | Developed rule-based NPC behavior and character mechanics in Unity.

BFactory Inc.

Dec. 2022 – Feb. 2023

Machine Learning Intern | Seoul, Republic of Korea

- Built an EleGANT-based makeup-transfer feature, including quality metrics, data pipelines, and fine-tuning; adapted diffusion models for personalized advertisement generation.

PUBLICATIONS & MANUSCRIPTS

*Equal contribution

- [M1] **OPAL: Operator Alignment for Unsupervised Graph Domain Adaptation**
Biniyam Aschalew Tolera, Akotet Yeshaw Tesema, Cheonwoo Lee, Dooho Lee, Aikta Arya, Jaemin Yoo
Submitted to NeurIPS 2026.
- [C1] **PULSE: Socially-Aware User Representation Modeling Toward Parameter-Efficient Graph Collaborative Filtering**
Doyun Choi*, Cheonwoo Lee*, Biniyam Aschalew Tolera, Taewook Ham, Chanyoung Park, Jaemin Yoo
ACM Web Conference (WWW), 2026. Oral presentation [Paper] [Code]
- [C2] **SelfReplay: Adapting Self-Supervised Sensory Models via Adaptive Meta-Task Replay**
Hyungjun Yoon, Jaehyun Kwak, Biniyam Aschalew Tolera, Gaole Dai, Mo Li, Taesik Gong, Kimin Lee, Sung-Ju Lee
ACM SenSys, 2025. [Paper]
- [C3] **By My Eyes: Grounding Multimodal Large Language Models with Sensor Data via Visual Prompting**
Hyungjun Yoon, Biniyam Aschalew Tolera, Taesik Gong, Kimin Lee, Sung-Ju Lee
Conference on Empirical Methods in Natural Language Processing (EMNLP Main), 2024. [Paper]
Qualcomm Innovation Fellowship Korea 2025
- [C4] **Leveraging Synthetic Data for Data-Free Knowledge Distillation**
Biniyam Aschalew Tolera, Bryan Nathanael Wijaya, Minhajur Rahman Chowdhury Mahim
Korean Software Conference (KSC), 2024. [Paper]

SELECTED PROJECTS

Synthetic Data for Zero-Shot Knowledge Distillation [C4] Spring 2024
KAIST CS570 course project [Code]

- Generated class-specific synthetic data by optimizing noise inputs to maximize teacher logits, enabling student training without access to the original training data.

Lang ReCAPTCHA Summer 2023
Junction Asia hackathon [Code] | Built a KoBERT-based CAPTCHA for low-resource language data collection and user authentication.

Fake Import Declaration Detection Spring 2023
3rd of 190+ participants, KAIST CS360 / Korean Customs Service competition; built model ensembles and data-processing pipelines.

HONORS & AWARDS

KAIST College of Engineering Leadership Award (Research Excellence)	Mar. 2025
Selected for outstanding research by a college-wide committee from department-nominated undergraduates.	
Daewoong AI and Big Data Global Scholarship	Mar. 2024
KAIST Full Scholarship	Mar. 2021
Hanseong-Sonjaehan Scholarship Award	Mar. 2021
MindPlus National Mathematics Olympiad Gold Medalist	Aug. 2019

TEACHING EXPERIENCE

Machine Learning, EAGLOPEN Course Instructor	Summer 2026
AI Foundation Models: Theory and Practice (EE499) Teaching Assistant	Spring 2026
AddisCoder Programming Summer Camp Teaching Assistant	Fall 2025
Introduction to Algorithms (CS300), KAIST Teaching Assistant	Fall 2022
Introduction to Programming (CS101), KAIST Teaching Assistant	Spring 2022

ACADEMIC SERVICE

Reviewer, ICML 2026 Graph Foundation Models (GFM) Workshop.
Student Volunteer, KDD 2026.

TECHNICAL SKILLS

Programming: Python, C, Java, SQL, MATLAB
Libraries and tools: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Git, Linux